Table 6.1. Functional groups and their basic parameters used in the Northern South China Sea ecosystem models for (a) the 1970s and (b) the 2000s states. Values in parentheses were estimated by the model.

(a) Basic parameters of the 1970s NSCS model

Group no.	Functional group	B	P/B	Q/B	EE
1	Phytoplankton	323	399	-	(0.035)
2	Benthic producer	153	11.89	-	(0.020)
3	Zooplankton	33.8	32	192	(0.052)
4	Jellyfish	(0.146)	5.00	20.0	0.950
5	Polychaetes	(3.421)	6.75	22.5	0.950
6	Echinoderms	3.065	1.20	3.58	(0.398)
7	Benthic crustaceans	2.649	5.65	26.9	(0.624)
8	Non-ceph molluscs	13.747	3.00	7.00	(0.383)
9	Sessile/other invertebrates	3.114	1.00	9.00	(0.845)
10	Shrimps	(0.422)	5.40	28.9	0.950
11	Crabs	(0.731)	3.00	12.0	0.950
12	Cephalopods	(0.465)	3.10	8.00	0.500
13	Threadfin bream (nemipterids)	1.04	0.74	8.10	(0.469)
14	Bigeyes (priacanthids)	0.318	1.21	11.3	(0.328)
15	Lizard fish (synodontids)	0.30	2.30	5.41	(0.241)
16	Juvenile Hairtail (trichiurids)	0.034	2.30	13.41	(0.329)
17	Adult hairtail (trichiurids)	0.0426	1.50	6.21	(0.327)
18	Pomfret (stromateids)	(0.065)	1.30	6.38	0.950
19	Snappers	(0.014)	1.34	8.98	0.950
20	Adult groupers	0.040	0.85	6.10	(0.375)
21	Croakers (≤ 30 cm)	0.289	2.36	11.28	(0.784)
22	Juvenile large croakers	0.0425	2.36	15.65	(0.913)
23	Croakers (> 30 cm)	0.095	1.43	6.23	(0.541)
24	Demesral fish (≤ 30 cm)	(1.541)	2.70	13.03	0.950
25	Juvenile demersal fish (> 30 cm)	0.112	2.60	15.46	(0.599)
26	Adult demersal fish (>30 cm)	0.195	1.44	6.21	(0.355)
27	Benthopelagic fish	(0.470)	3.00	15.0	0.950
28	Melon seed	0.114	2.24	24.7	(0.964)
29	Pelagic fish (≤ 30 cm)	(1.050)	2.87	12.22	0.950
30	Juvenile large pelagic fish	0.118	2.87	14.37	(0.62)
31	Pelagic fish (> 30 cm)	0.158	0.90	6.28	(0.283)
32	Demersal sharks and rays	0.04	1.26	6.30	(0.364)
33	Pelagic sharks and rays	(0.028)	0.39	1.95	0.500
34	Seabirds	0.0022	0.06	67.76	(0.005)
35	Pinnipeds	0.0046	0.045	14.77	(0.679)
36	Other mammals	0.0158	0.112	10.52	(0.068)
37	Marine turtles	0.0002	0.100	3.50	(0.503)
38	Detritus	100	-	-	(0.017)

(b) Basic parameters of the 2000s NSCS model

Group no.	Functional group	B	P/B	Q/B	EE
1	Phytoplankton	323	398		(0.010)
2	Benthic producer	153	11.89		(0.010)
3	Zooplankton	9.0	32.0	192	(0.306)
4	Jellyfish	1.53	5.00	20.0	(0.520)
5	Polychaetes	2.24	6.75	22.5	(0.673)
6	Echinoderms	1.98	1.20	3.58	(0.444)
7	Benthic crustaceans	1.43	5.65	26.9	(0.617)
8	Non-ceph molluscs	2.68	3.50	11.7	(0.951)
9	Sessile/other invertebrates	2.61	1.00	9.00	(0.575)
10	Shrimps	(0.194)	7.60	28.94	(0.950)
11	Crabs	(0.368)	3.00	12.0	(0.950)
12	Cephalopods	0.68	3.10	8.00	(0.393)
13	Threadfin bream (nemipterids)	0.26	3.08	15.4	(0.847)
14	Bigeyes (priacanthids)	0.13	3.33	11.3	(0.550)
15	Lizard fish (synodontids)	0.032	1.60	5.407	(0.658)
16	Juvenile Hairtail (trichiurids)	0.015	3.08	14.894	(0.749)
17	Adult hairtail (trichiurids)	0.012	1.47	6.207	(0.545)
18	Pomfret (stromateids)	0.108	3.03	(15.15)	0.950
19	Snappers	(0.0013)	1.75	8.984	0.950
20	Adult groupers	(0.0064)	1.75	6.10	0.950
21	Croakers (≤ 30 cm)	0.07	3.30	11.276	(0.958)
22	Juvenile large croakers	0.04	3.30	16.366	(0.564)
23	Croakers (> 30 cm)	0.0094	1.43	6.232	(0.587)
24	Demesral fish (≤ 30 cm)	(0.316)	4.70	23.5	0.950
25	Juvenile demersal fish (> 30 cm)	0.143	3.50	16.144	(0.722)
26	Adult demersal fish (>30 cm)	0.021	2.10	6.207	(0.747)
27	Benthopelagic fish	0.922	3.08	15.42	(0.479)
28	Melon seed	0.070	2.41	24.0	(0.994)
29	Pelagic fish (≤ 30 cm)	1.772	4.26	17.04	(0.740)
30	Juvenile large pelagic fish	0.289	4.26	16.12	(0.622)
31	Pelagic fish (> 30 cm)	0.079	1.40	6.27	(0.759)
32	Demersal sharks and rays	0.001	1.20	6	(0.867)
33	Pelagic sharks and rays	(0.0011)	0.68	3.4	0.950
34	Seabirds	0.0022	0.06	67.759	(0.046)
35	Pinnipeds	0.0046	0.045	14.768	(0.290)
36	Other mammals	0.0158	0.112	10.523	(0.034)
37	Marine turtles	0.0002	0.10	3.50	(0.300)
38	Detritus	100	-	-	(0.005)

Basic model input parameters were estimated from government surveys, published literature, empirical equations and global databases (see Table 6.1 and Appendix 6.1 for detailed descriptions of parameter estimations). Biomasses of most commercially important groups in the 2000s model were estimated based on trawl and acoustic surveys conducted by the South China Sea Fisheries Research Institute (Jia *et al.* 2004). In the 1970s model, their biomasses were back-calculated from the observed changes in relative abundance between the 1970s and 2000s as reported in published literature and government reports (see Appendix 6.1 for details). P/B ratios were based on mortality estimates from length-based studies and empirical equations (Pauly 1980). Q/B ratios were estimated from empirical equations (Palomares & Pauly 1998). Diet compositions were based on local surveys (Xu *et al.* 1994) and the information available from FishBase (Froese & Pauly 2004) (Appendix 6.2).

The 2000s ecosystem was exploited by six fleets categorized by their fishing gears: pair and stern trawls, shrimp trawl, purse seine, hook and line, gillnet and 'others'; all fishing gears were aggregated into one fishing fleet in the 1970s model. The specifications of fishing sectors in the 2000s model facilitated the identification of optimal fleet configurations through dynamic ecosystem simulations (Chapter 8). However, the limited resolution of catch data by fishing fleets in the 1970s did not allow segregation of catches by fishing fleet types for that period. On the other hand, this did not affect the comparison of ecosystem structures between the 1970s and 2000s as the models only accounted for the total catches by all fishing fleets.

Catches in the 1970s and 2000s model were based on landings statistics reported by the PRC government. However, the PRC landings data have been suggested to be largely over-estimated (Watson & Pauly 2001). The *Sea Around Us* Project (SAUP) provided (www.seaaroundus.org) catch estimates for China that had been adjusted downward based on a meta-analysis of global fisheries catches (Watson & Pauly 2001). Thus the SAUP data were also used to estimate the catches in the two models (Appendix 6.1).

Catches by fishing sectors and functional groups were estimated from national and regional fisheries statistics. Catches from the six fishing sectors in the 2000s model were obtained from the PRC fisheries statistics. The national statistics did not report catch composition by fishing fleets. However, such data were available for Hong Kong fisheries and as fishing vessels and gears used in Hong Kong are typical of those in the NSCS region, species composition by fleets was prorated according to the relative catch compositions of Hong Kong fishing fleets (Pitcher *et al.* 1998) (Table 6.2).

The initial input parameters did not result in a model that met the mass-balance criteria. Thus the input parameters were adjusted iteratively until the model achieved mass-balance i.e., the values of EE of all functional groups were below 1 (Appendix 6.1). The diet composition matrix was the primary parameter adjusted because it was relatively more uncertain than the other input parameters. However, changing the diet composition matrix alone was not enough for the models to achieve mass-balance. Thus the P/B, Q/B and biomass inputs were adjusted based on the relative accuracy of the data sources and the available information. The procedures for defining the relative accuracy of the input parameters are described in a later section.

Table 6.2. Estimated fishery catch ($t \cdot km^{-2}$) by functional groups and gear types in the 1970s and 2000s models. PSt – pair and stern trawl; ShT – shrimp trawl; PS – purse seine; H&L – hook and line; GN – gillnet; Others – other fishing gears; inverts – invertebrates; Ju. – juvenile; Ad. – adult; Dem. – demersal.

Functional groups	Catches by functional groups ($t \cdot km^{-2}$)							
	1970s model	2000s model						
	Total	PSt	ShT	PS	H&L	GN	Others	Total
Phytoplankton	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.000
Benthic producer	0.0056	0.0000	0.0000	0.0000	0.0000	0.0000	0.0056	0.006
Zooplankton	0.0100	0.0000	0.0000	0.0065	0.0000	0.0000	0.0883	0.095
Jellyfish	0.0012	0.0126	0.0000	0.0314	0.0000	0.0000	0.0000	0.044
Polychaetes	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.000
Echinoderms	0.0039	0.0001	0.0019	0.0000	0.0000	0.0000	0.0001	0.002
Benthic crustaceans	0.0019	0.0003	0.0388	0.0000	0.0000	0.0000	0.0000	0.039
Non-ceph molluscs	0.0056	0.0763	0.1800	0.0000	0.0000	0.0000	0.5060	0.762
Sessile/other inverts	0.0011	0.0001	0.0029	0.0000	0.0000	0.0000	0.0000	0.003
Shrimps	0.0350	0.0801	0.5760	0.0001	0.0000	0.0000	0.0226	0.679
Crabs	0.0100	0.0400	0.1000	0.0000	0.0000	0.0421	0.0176	0.200
Cephalopods	0.0244	0.1530	0.0243	0.0627	0.0009	0.0000	0.0324	0.273
Threadfin bream	0.0440	0.4590	0.0089	0.0000	0.0651	0.0994	0.0246	0.657
Bigeyes	0.0350	0.1420	0.0029	0.0000	0.0210	0.0321	0.0080	0.206
Lizard fish	0.0840	0.0012	0.0015	0.0000	0.0000	0.0166	0.0041	0.023

Table 6.2 Con't.

Functional groups	Catches by functional groups (t·km^{-2}) 1970s model	2000s model						
	Total	PSt	ShT	PS	H&L	GN	Others	Total
Juv. Hairtail	0.0038	0.0206	0.0005	0.0001	0.0000	0.0054	0.0014	0.028
Ad. hairtail	0.0152	0.0047	0.0000	0.0001	0.0009	0.0012	0.0003	0.007
Pomfret	0.0053	0.0809	0.0076	0.0154	0.0299	0.0843	0.0209	0.239
Snappers	0.0053	0.0003	0.0001	0.0001	0.0002	0.0003	0.0001	0.001
Ad. groupers	0.0029	0.0013	0.0003	0.0000	0.0012	0.0049	0.0012	0.009
Croakers (≤ 30 cm)	0.0160	0.0159	0.0117	0.0000	0.0003	0.0027	0.0045	0.035
Juv. large croakers	0.0110	0.0371	0.0033	0.0000	0.0060	0.0168	0.0078	0.071
Croakers (> 30 cm)	0.0440	0.0046	0.0000	0.0000	0.0001	0.0032	0.0001	0.008
Dem. fish (≤ 30 cm)	0.0902	0.0085	0.0760	0.0100	0.0055	0.0262	0.0526	0.179
Juv. Dem. fish (> 30 cm)	0.0288	0.0230	0.2070	0.0100	0.0000	0.0613	0.0152	0.317
Ad. Dem. fish (>30 cm)	0.1150	0.0285	0.0000	0.0000	0.0023	0.0035	0.0009	0.035
Benthopelagic fish	0.0303	0.3420	0.0278	0.0607	0.0010	0.0580	0.1130	0.603
Melon seed	0.0057	0.0284	0.0023	0.0050	0.0000	0.0048	0.0094	0.050
Pelagic fish (≤ 30 cm)	0.1460	0.5120	0.0000	1.0860	0.0000	0.1030	0.6440	2.345
Juv. large pelagic fish	0.0096	0.2080	0.0000	0.0829	0.0000	0.3750	0.0725	0.738
Pelagic fish (> 30 cm)	0.0380	0.0185	0.0000	0.0074	0.0165	0.0333	0.0064	0.082
Dem. sharks and rays	0.0158	0.0006	0.0002	0.0000	0.0000	0.0002	0.0000	0.001
Pelagic sharks and rays	0.0051	0.0002	0.0000	0.0000	0.0002	0.0003	0.0000	0.001
Seabirds	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.000
Pinnipeds	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.000
Other mammals	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.000
Marine turtles	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.000
Detritus	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.000
Total	0.8500	2.2990	1.2740	1.3780	0.1510	0.9750	1.6590	7.736

6.2.2. Uncertainty and sensitivity analysis

Uncertainties of the input parameters were specified under a 'pedigree' in the Ecopath with Ecosim package (Christensen *et al.* 2004). The 'pedigree' is a matrix that allowed systematic categorisation of the reliability of the input parameters – specifically the biomass (B); production to biomass ratio (P/B); and the consumption to biomass ratio (Q/B); the diet composition matrix and the catch of each functional group. In the pedigree routine, a coded statement categorizing the origin (data type and associated uncertainty) of a given input was given to each of these parameters. Inputs were rated based on how they had been derived: local data, other locations, 'best guesses', empirical relationships, other Ecopath models, or estimated by the current model. Associated with each of these categories was an index of data quality which ranged from 0 to 1, with 0 being the lowest quality (estimated by the model while solving the mass-balance equations) while 1 being the highest quality (e.g. data from a quantitative study conducted in the study area). By summing across these pedigrees, an index (P) of the overall quality of the input information in Ecopath can be calculated: